

CHD-CARE Online Demo Operation Guide

Collaborative AI for Reliable Pediatric Congenital Heart Disease Diagnosis Across Tiered Hospitals

Manuscript ID	NCOMMS-26-037516
Demo URL	http://124.156.173.246:8000/
Repository	https://github.com/strewberrr/CHD-CARE
Access note	Account credentials are not included in this public guide and should be provided separately when required.

Purpose

This online research demo is provided to help editors, reviewers, and academic users inspect the CHD-CARE inference and visualization workflow without deploying the code locally. The platform supports built-in example cases or upload of de-identified color Doppler echocardiography videos for case-level inference.

The demo returns four-class diagnostic probabilities for Normal, VSD, ASD, and PDA, and provides visual guidance including original videos, heatmap overlays, key-frame-oriented visualization, and defect-region bounding-box displays when applicable.

Important: This demo is for peer-review and academic evaluation only. It is not a certified medical device and must not be used for clinical diagnosis, treatment decisions, or patient management. Users should upload only de-identified data and should not upload any protected health information.

Quick start

1	Open the demo URL and sign in if an account is required.
2	Enter the diagnosis mode. The platform will show the task upload interface.
3	Upload one task folder, one case folder, or multiple video files from a single case. The platform does not support selecting multiple folders simultaneously; upload folders one at a time.
4	After upload, the AI model will automatically process the task. Completed tasks can be opened from the task status page.
5	Review each case using Original, Heatmap, and Bounding Box modes. The right panel shows case-level probabilities for Normal, VSD, ASD, and PDA.
6	In Heatmap or Bounding Box mode, click a video to enlarge it. Click the Locate button to jump to and pause at the model-selected key frame.
7	Optionally select a clinical judgment for each case and submit all results.

Input organization

The demo accepts one task folder, one case folder, or multiple video files belonging to one case. The platform automatically organizes uploaded videos into the expected task/case structure. Each case can contain up to five videos; additional videos may be ignored or truncated by the platform.

Folder uploads must be performed one folder at a time. If users need to evaluate multiple folders, upload and process each folder separately. Supported video formats include .mp4, .avi, .mov, and .mkv.

Example structure: Task1/patient1/video1.mp4, video2.mp4, ... up to five videos per case.

Reviewing model outputs

Original mode

Displays the uploaded color Doppler echocardiography videos. Videos loop automatically. Users may click a video to enlarge it, pause or resume playback, and use the progress bar to inspect specific frames.

Heatmap mode

Displays heatmap overlays on the model-selected or displayed video frames. This mode helps users inspect the spatial evidence used by the model. In the enlarged view, the Locate button jumps to the model-selected key frame and pauses playback at that frame.

Bounding Box mode

Displays defect-region localization boxes when the model predicts a positive finding. For cases predicted as negative or normal, the platform may display "Negative, no detection box" instead of forcing a localization box.

Key-frame locate function

When reviewing a video in Heatmap or Bounding Box mode, click the video to open the enlarged view. The top bar shows the model-selected key-frame index. Click the Locate button to automatically navigate to this key frame and freeze the video at the model-identified diagnostically informative moment.

Diagnostic panel

The right-side panel reports model confidence for four categories: Normal, VSD, ASD, and PDA. Users may optionally select a clinical judgment from the same four categories. This manual selection is intended for review or reader-study workflows and does not change the AI-generated model probabilities.

Downloading results

After completing a task, users may download the result files from the task result panel. The available outputs may include a physician or user judgment report in JSON format and a model-output ZIP archive.

File	Description
Judgment report JSON	Contains submitted case-level judgments, task information, timestamps, and viewing time records when applicable.
Model-output ZIP	Contains model-generated outputs for each case, including processed video files, per-video JSON files, key-frame information, heatmap or localization metadata, and case-level confidence scores.

Notes for public distribution

- This guide does not include any username or password. Access credentials, if required, should be provided separately through a non-public channel.
- Do not upload protected health information, patient identifiers, or non-de-identified clinical videos to the public demo.
- The demo is intended to demonstrate CHD-CARE inference and visual-guidance workflow, not to reproduce all quantitative results in the manuscript.
- Full reproduction of manuscript-level results requires controlled access to the de-identified study dataset according to the Data Availability statement.